Semih Cayci

ADDRESS: Pontdriesch 14-16
52062 Aachen, Germany
PHONE: +49 241 80-8097828
EMAIL: cayci@mathc.rwth-aachen.de
WEB: https://semihcayci.github.io

RESEARCH INTERESTS

Mathematical foundations of artificial intelligence: deep learning and reinforcement learning, optimization theory, applied stochastic analysis

EMPLOYMENT HISTORY

10/2022-PRESENT — Tenure-Track Assistant Professor for Mathematics of Machine Learning
RWTH AACHEN UNIVERSITY, AACHEN, GERMANY
Department of Mathematics
Positive interim evaluation (2025)

08/2020-08/2022 — NSF TRIPODS Postdoctoral Fellow
UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN, URBANA, IL, USA
Coordinated Science Laboratory
Illinois Institute of Data Science and Dynamical Systems (IDS2)
HOSTS: Prof. Niao He (ETH Zurich) and Prof. R. Srikant (UIUC)

EDUCATION

2014-2020 — Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering
The Ohio State University, COLUMBUS, OH, USA
DISSERTATION: Online Learning for Optimal Control of Communication and Computing Systems
ADVISOR: Prof. Atilla Eryilmaz

2010-2013 — Master of Science (M.Sc.) in Electrical and Electronics Engineering
Bilkent University, ANKARA, TURKEY
THESIS: Lossless Data Compression with Polar Codes
ADVISORS: Prof. Erdal Arıkan and Prof. Orhan Arıkan

2005-2010 — Bachelor of Science (B.Sc.) in Electrical and Electronics Engineering
Bogazici University, ISTANBUL, TURKEY

RESEARCH EXPERIENCE

2020-2022 — **NSF TRIPODS Postdoctoral Fellow**
UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN, URBANA, IL, USA
Coordinated Science Laboratory
ADVISORS: Prof. Niao He (ETH Zurich) and Prof. R. Srikant (UIUC)
PROJECTS: Theoretical foundations of reinforcement learning

2014-2020 — **Graduate Research Assistant**
THE OHIO STATE UNIVERSITY, COLUMBUS, OH, USA
Department of Electrical and Computer Engineering
ADVISOR: Prof. Atilla Eryilmaz
PROJECTS: Knapsack-constrained bandits and online machine learning
Learning-based control of networked systems

06/2017-09/2017 — Intern Scientist
MITSUBISHI ELECTRIC RESEARCH LABS (MERL), CAMBRIDGE, MA, USA
HOST: Toshiaki Koike-Akino
PROJECTS: Polar coding for optical communications

2010-2014 **Graduate Research Assistant**
BILKENT UNIVERSITY, ANKARA, TURKEY
ADVISORS: Prof. Erdal Arıkan and Prof. Orhan Arıkan
PROJECTS: Polar source coding for q-ary alphabets

2007-2010 **Undergraduate Researcher**
BOGAZICI UNIVERSITY, ISTANBUL, TURKEY
Intelligent Systems Laboratory
ADVISOR: Prof. H. Isil Bozma
PROJECTS: Scene and object recognition for autonomous robot navigation
Motion planning for autonomous robots

PUBLICATIONS

5.1 Selected Publications

[1] **S. Cayci**, "Geometric Convergence of Gauss-Newton for Neural Networks: Riemannian Geometry and Adaptive Damping", ICML 2026.

[2] **S. Cayci**, N. He, and R. Srikant, "Convergence of Entropy-Regularized Natural Policy Gradient with Linear Function Approximation," SIAM Journal on Optimization 34.3, pp. 2729-2755, 2024.

[3] **S. Cayci**, "Non-Asymptotic Generalization and Optimization Bounds for Stochastic Gauss-Newton in Deep Neural Networks", AISTATS 2026.

[4] A. B. Yardim, **S. Cayci**, M. Geist, N. He, "Policy Mirror Ascent for Efficient and Independent Learning in Mean Field Game," International Conference on Machine Learning (ICML), 2023.

[5] **S. Cayci**, A. Eryilmaz, "Convergence of Gradient Descent for Recurrent Neural Networks: A Nonasymptotic Analysis," SIAM Journal on Mathematics of Data Science 7.2, pp. 826-854, 2025.

[6] **S. Cayci**, N. He, R. Srikant, "Finite-Time Analysis of Natural Actor-Critic for POMDPs," SIAM Journal on Mathematics of Data Science, 6(4), 869-896, 2024.

[7] **S. Cayci**, A. Eryilmaz, "Provably Robust Temporal Difference Learning for Heavy-Tailed Rewards," Advances in Neural Information Processing Systems (NeurIPS) 36, 2023.

[8] **S. Cayci**, A. Eryilmaz, "Recurrent Natural Policy Gradient for POMDPs," Transactions on Machine Learning Research (TMLR), https://openreview.net/forum?id=6G01e0vgIf, 2025.

[9] J. Müller, **S. Cayci**, G. Montúfar, "Fisher–Rao Gradient Flows of Linear Programs and State-Action Natural Policy Gradients," SIAM Journal on Optimization 35.2, pp. 1060-1088, 2025.

[10] **S. Cayci**, S. Gupta, A. Eryilmaz, "Group-Fair Online Allocation in Continuous Time", Advances in Neural Information Processing Systems (NeurIPS) 33, pages 13750–13761, 2020.

5.2 Journal Papers

[J1] N. Oberweis, **S. Cayci**, "Convergence of Stochastic Gradient Langevin Dynamics in the Lazy Training Regime," Transactions on Machine Learning Research, https://openreview.net/forum?id=3s037DKmLo, 2026.

[J2] J. Müller, **S. Cayci**, "Optimal Rates of Convergence for Entropy Regularization in Discounted Markov Decision Processes," Information and Inference: A Journal of the IMA, Volume 15, Issue 1, March 2026, iaaf034, 2026.

[J3] **S. Cayci**, A. Eryilmaz, "Convergence of Gradient Descent for Recurrent Neural Networks: A Nonasymptotic Analysis," SIAM Journal on Mathematics of Data Science 7.2, pp. 826-854, 2025.

[J4] **S. Cayci**, A. Eryilmaz, "Recurrent Natural Policy Gradient for POMDPs," Transactions on Machine Learning Research (TMLR), https://openreview.net/forum?id=6G01e0vgIf, 2025.

[J5] J. Müller, **S. Cayci**, G. Montúfar, "Fisher–Rao Gradient Flows of Linear Programs and State-Action Natural Policy Gradients," SIAM Journal on Optimization 35.2, pp. 1060-1088, 2025.

[J6] **S. Cayci**, Y. Zheng, A. Eryilmaz, "Budgeted-Bandits with Controlled Restarts with Applications in Learning and Computing", Transactions on Machine Learning Research (TMLR), https://openreview.net/forum?id=lvb5qDAa4B, 2025.

[J7] B. Yardim, **S. Cayci**, N. He, "A Variational Inequality Approach to Independent Learning in Static Mean-Field Games," ACM/IMS Journal on Data Science 2.2, 2025.

[J8] **S. Cayci**, N. He, and R. Srikant, "Convergence of Entropy-Regularized Natural Policy Gradient with Linear Function Approximation," SIAM Journal on Optimization 34.3, pp. 2729-2755, 2024.

[J9] **S. Cayci**, N. He, R. Srikant, "Finite-Time Analysis of Natural Actor-Critic for POMDPs," SIAM Journal on Mathematics of Data Science, 6(4), 869-896, 2024.

[J10] **S. Cayci**, N. He, R. Srikant, "Finite-Time Analysis of Entropy-Regularized Neural Natural Actor-Critic Algorithm," Transactions on Machine Learning Research, https://openreview.net/forum?id=BkEqk7pS1I, 2024.

[J11] Y. Zheng, **S. Cayci**, A. Eryilmaz, "Fast Online Learning of Vulnerabilities for Networks With Propagating Failures," IEEE/ACM Transactions on Networking, 2024.

[J12] **S. Cayci**, S. Satpathi, N. He, R. Srikant, "Sample Complexity and Overparameterization Bounds for Temporal Difference Learning with Neural Network Approximation," IEEE Transactions on Automatic Control, 68.5, pp. 2891-2905, 2023.

[J13] **S. Cayci**, A. Eryilmaz, R. Srikant, "Learning to Control Renewal Processes with Bandit Feedback", Proceedings of the ACM on Measurement and Analysis of Computing Systems 3.2, Article 43, 2019.

[J14] **S. Cayci**, A. Eryilmaz, "Optimal Learning for Dynamic Coding in Deadline-Constrained Multi-Channel Networks," IEEE/ACM Transactions on Networking, 27.3, pp. 1043-1054, 2019.

[J15] H. Song, S. Singh, **S. Cayci**, M. Torrilhon, "Machine-learned R13 moment closures for shock-dominated rarefied gas flows," Computers & Fluids, 107260, 2026.

5.3 Conference Proceedings

[C1] **S. Cayci**, "Geometric Convergence of GaussNewton for Neural Networks: Riemannian Geometry and Adaptive Damping", ICML 2026 (acceptance rate: 26.6%)

[C2] **S. Cayci**, "Non-Asymptotic Generalization and Optimization Bounds for Stochastic Gauss-Newton in Deep Neural Networks", AISTATS 2026 (acceptance rate: 28%)

[C3] E. Arda, **S. Cayci**, A. Eryilmaz, "Finite-Time Analysis of Gradient Descent for Shallow Transformers", AISTATS 2026 (acceptance rate: 28%)

[C4] **S. Cayci**, A. Eryilmaz, "Provably Robust Temporal Difference Learning for Heavy-Tailed Rewards," Proceedings of Neural Information Processing Systems (NeurIPS) 36, 2023. (acceptance rate: 26.1%)

[C5] A. B. Yardim, **S. Cayci**, M. Geist, N. He, "Policy Mirror Ascent for Efficient and Independent Learning in Mean Field Game," International Conference on Machine Learning (ICML), 2023. (acceptance rate: 27.9%)

[C6] **S. Cayci**, S. Gupta, A. Eryilmaz, "Group-Fair Online Allocation in Continuous Time", Proceedings of Neural Information Processing Systems (NeurIPS) 33, 2020. (acceptance rate: 20.1%)

[C7] **S. Cayci**, Y. Zheng, A. Eryilmaz, "A Lyapunov-Based Methodology for Constrained Optimization with Bandit Feedback," Proceedings of AAAI Conference on Artificial Intelligence (AAAI), 2022. (acceptance rate: 15.0%)

[C8] **S. Cayci**, A. Eryilmaz, R. Srikant, "Budget-Constrained Bandits over General Cost and Reward Distributions", Proceedings of International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 108:4388-4398, 2020. (acceptance rate: 28.0%)

[C9] **S. Cayci**, A. Eryilmaz, R. Srikant, "Learning to Control Renewal Processes with Bandit Feedback", ACM SIGMETRICS, June 2019. (acceptance rate: 16.0%)

[C10] **S. Cayci** and A. Eryilmaz, "Learning for Serving Deadline-Constrained Traffic in Multi-Channel Wireless Networks," 15th IEEE International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), Paris, 2017.

[C11] **S. Cayci**, O. Arikan, "Lossless Polar Compression of q-ary Sources," IEEE International Symposium on Information Theory Proceedings (ISIT), 2013, pp. 1132-1136, 7-12 July 2013

[C12] **S. Cayci**, O. Arikan, E. Arikan, "Polar Code Construction for Non-Binary Source Alphabets", IEEE 20th Signal Processing and Communications Applications Conference, pp. 1-4, 18-20 April 2012

[C13] **S. Cayci**, A. Eryilmaz, “On the Multi-Channel Capacity Gains of Millimeter-Wave Communication," 2016 IEEE Global Communications Conference (GLOBECOM), Washington D.C., 2016

[C14] **S. Cayci**, T. Koike-Akino, Y. Wang, “Nonbinary Polar Coding for Multilevel Modulation", Optical Fiber Communication Conference, pp. W3H-4. Optical Society of America, 2019.

[C15] T. Koike-Akino, Y. Wang, **S. Cayci**, D. S. Millar, K. Kojima, K. Parsons, “Hardware-Efficient Quantized Polar Decoding with Optimized Lookup Table", in IEEE 2019 24th OptoElectronics and Communications Conference (OECC), pp. 1-3, 2019.

MANUSCRIPTS UNDER REVIEW

[M1] R. Zolnouri, **S. Cayci**, “Preference-Based Reward Learning in POMDPs with Inexact Dynamics", under review, 2026.

SELECTED TALKS

Lectures

- “Policy Gradient Methods", Reinforcement Learning Summer School 2026, Milan, Italy, June 5, 2026.
- “Theoretical Aspects of Deep Learning", SFB 1481 Spring School, Kloster Steinfeld, May 21-25, 2023.

Selected Talks

- "Policy Optimization for POMDPs: Sample Complexity and Convergence Guarantees", Institute for Systems Theory and Automatic Control at the University of Stuttgart, Stuttgart, Germany, November 17, 2026.
- “Geometric Convergence of Gauss-Newton in Deep Learning: Riemannian Geometry and Adaptive Damping", Swiss Optimization Symposium, Ascona, Switzerland, August 27, 2026.
- “Reinforcement Learning for Partially Observable Mean-Field Games", ECC 2026 Workshop on Learning in Dynamic Games under Partial Information, July 7, 2026.
- “Convergence of Gauss-Newton in the Lazy Training Regime: A Riemannian Optimization Perspective", TU Hamburg, September 23, 2025.
- “A Neural Tangent Kernel Analysis of Adaptively-Regularized GaussNewton in Deep Learning", Mini-Workshop on Scientific Machine Learning and Uncertainty Quantification in Aachen, May 8, 2025.
- “Recurrent Natural Policy Gradient for POMDPs", McGill University, January 24, 2025.
- “Convergence of Recurrent Neural Networks Trained with Gradient Descent and Temporal Difference Learning", ODI Group at ETH Zurich, May 10, 2024.
- “A Non-Asymptotic Analysis of Gradient Descent for Recurrent Neural Networks", Mathematics of Machine Learning Seminars, Max Planck Institute and University of California at Los Angeles, April 11, 2024.
- “Sample Efficiency and Robustness of Natural Actor-Critic Methods with Function Approximation", invited speaker at the Workshop on Reinforcement Learning, University of Mannheim, January 12, 2024.
- “Towards Robust and Sample-Efficient Reinforcement Learning with Policy Optimization", Airbus Operations GmbH, Hamburg, March 5, 2024.
- “Sample Complexity and Overparameterization Bounds for Neural Reinforcement Learning", German Mathematical Society Annual Meeting, Ilmenau, Germany on September 25, 2023.
- “Convergence of Natural Actor-Critic Algorithm with Linear and Neural Network Approximation", Foundations of Computational Mathematics 2023, Paris, France on June 16, 2023.
- “Finite-Time Analysis of Natural Actor-Critic with Neural Network Approximation ", SIAM Conference on Mathematics of Data Science 2022 (MDS22) on September 27, 2022.
- “Convergence of Natural Actor-Critic Algorithm with Linear and Neural Network Approximation", Kick-Off Workshop for Sparsity and Singular Structures (SFB 1481), RWTH Aachen University on September 21, 2022.
- “Provably Scalable and Sample-Efficient Reinforcement Learning Using Policy Optimization", University of Connecticut, Department of Computer Science and Engineering in Storrs, CT, USA on April 19, 2022.

- “Provably Scalable and Sample-Efficient Reinforcement Learning Using Policy Optimization", Washington University, Department of Computer Science and Engineering in St. Louis, MO, USA on February 25, 2022.
- “Understanding Policy Optimization for Large-Scale Decision-Making Problems", Clemson University, Department of Industrial Engineering in Clemson, SC, USA on February 21, 2022.
- “Convergence of Reinforcement Learning Algorithms with Function Approximation", Bilkent University, Department of Computer Engineering on November 19, 2021.
- “On the Convergence of Entropy-Regularized Natural Policy Gradient with Function Approximation", INFORMS Annual Meeting 2021 on October 25, 2021.
- “Convergence of Entropy-Regularized Natural Policy Gradient with Function Approximation", Microsoft Research New York on September 23, 2021.
- “Faster Reinforcement Learning Algorithms with Function Approximation", NSF TRIPODS Annual PI Meeting 2021 on June 11, 2021.
- “Budget-Constrained Learning and Optimization with Bandit Feedback", Illinois Institute for Data Science and Dynamical Systems Colloquium, September 12, 2020.
- “Budget-Constrained Learning and Optimization with Bandit Feedback", Georgia Institute of Technology, Algorithms and Randomness Center Colloquium, February 3, 2020.
- “Learning to Control Renewal Processes with Bandit Feedback" at Midwest Machine Learning Symposium 2019 (Spotlight talk) on June 6, 2019.

PROFESSIONAL SERVICE

Organization committee: ACM SIGMETRICS 2027 (publicity chair), ACM SIGMETRICS 2026 (publicity chair), IEEE WiOpt 2026 (publicity co-chair)

Memberships: ELLIS NRW Unit, ELLIS - European Lab for Learning and Intelligent Systems, RWTH Center for Artificial Intelligence, Collaborative Research Center: SFB 1481 Sparsity and Singular Structures, EDDy Research and Teaching Group

University service: Member of the hiring committees for a W2 Professorship in Mathematical Optimization (2026) and a W2 Professorship in Industrial Mathematics (2025), RWTH Aachen University

Program committee: IEEE INFOCOM 2027, AAAI 2027, AAAI 2026, ACM MobiHoc 2025, Conference on Learning Theory (COLT) 2022, IEEE WiOpt 2021

Reviewer: ICML, NeurIPS, AISTATS, ICLR, COLT, Journal of Machine Learning Research, SIAM Journal on Mathematics of Data Science, SIAM Journal on Control and Optimization, Mathematics of Operations Research, IEEE/ACM Transactions on Networking, IEEE ISIT, IEEE CDC, IEEE Transactions on Network Science and Engineering, Performance Evaluation, IEEE WiOpt, ACM MobiHoc

TEACHING EXPERIENCE

RWTH Aachen	**Professor** Mathematical Foundations of Machine Learning, Wi 2025 Mathematical Foundations of Reinforcement Learning, Wi 2022, Su 2025 Mathematical Foundations of Deep Learning, Wi 2023, Su 2026 Stochastic Analysis, Wi 2024 Seminar on Deep Learning Theory, Su 2023
Ohio State	**Teaching Assistant** ECE 7100: Network Optimization and Control Sp 2017 ECE 7001: Detection and Estimation Theory, Sp 2018
Bilkent University	**Teaching Assistant** EEE 431: Telecommunications I, Au 2011, Au 2012 EEE 432: Telecommunications II, Sp 2012, Sp 2013 EEE 525: Advanced Digital Signal Processing, Sp 2013 MATH 255: Probability, Au 2013 EEE 521: Radar Signal Processing, Sp 2014 MATH 241: Engineering Mathematics I, Au 2010 MATH 242: Engineering Mathematics II, Sp 2011

MENTORING

- **Post-doc:** Johannes Müller, 2023-2024 (next position: TU Berlin)
- **PhD:** Reza Zolnouri

MSc students:

- Xinyi Sheng (next position: PhD student at Aalto University)
- Jonas Nießen (next position: PhD student at RWTH Aachen)
- Xingguang Zhou
- Yang Yang
- Reza Zolnouri (next position: PhD student at RWTH Aachen)
- Jiaan Xing
- Steve Meiter
- Mariia Krasilnikova

GRANTS

- **Mathematical and Practical Properties of Policy-Based Reinforcement Learning (OPSF854)**, ERS at RWTH Aachen University, co-PIs: Semih Cayci and Sebastian Trimpe (Total amount: 79,000 EUR)
- **GPU-Accelerated Benchmarking of Learning-Based Routing under Heavy Tails**, NVIDIA Academic Grant, PI: Semih Cayci (2x NVIDIA DGX Spark Supercomputers)